1.3 How energy is generated and stored

1.3.1 Sources, generation and storage of energy

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Table 1.5 Non-renewable energy sources

Oil	Crude oil is extracted from the Earth and refined into liquid fuels such as petrol. It can also be used to generate electricity in power stations.
Gas	Extracted through drilling and piped through the national grid to houses and factories. It can also be used to generate electricity in power stations.
Coal	Mined from the ground and burnt in power stations to generate electricity.

Issues surrounding the use of fossil fuels:

- Fossil fuels have high energy density – they hold a lot of chemical energy per kilogram of fuel – making them ideal for transportation.
- Gases such as carbon dioxide and pollutants such as sulphur dioxide are emitted when fossil fuels are burnt. This can cause breathing problems and contributes to global warming.
- Fossil fuels cannot be replaced and will eventually run out.

Table 1.6 Renewable energy sources

Wind	A wind turbine extracts energy from the wind. The blades are connected to a generator that produces electricity.
Solar	Photovoltaic (PV) panels produce electricity when exposed to the sunlight.
Hydroelectric	Dams house large turbines that trap water. When the water is released, the pressure turns the turbines and generate electricity.
Biofuels – biodiesel and biomass	Wood not used in the timber industry is chipped and used as fuel (biodiesel) instead of burning coal. In some biomass schemes plants such as soy are grown to produce materials that can be processed into biofuels.
Tidal	Energy is extracted from the rise and fall of the tide. Large turbines can be placed in areas where there is high tidal movement.

Table 1.7 Advantages and disadvantages of renewable energy sources

Advantages	Disadvantages
Renewable energy sources are non-polluting and considered better for the environment.	Initial outlay for the equipment needed for renewable energy is expensive (it produces free energy following installation, however).
Although biomass fuels release carbon dioxide as they are burned, trees are replanted, which absorb carbon dioxide as they grow. The process is classed as **carbon neutral**.	Wind and solar power depend on weather conditions and therefore cannot be relied on.
Manufacturers are installing equipment to recover waste energy to heat their offices. This will reduce energy bills, but is also a more ethical approach to construction.	In order to build dams to generate hydroelectric power, valleys in rural areas must be flooded. This can damage the natural habitat for wildlife.

Carbon neutral: no net release of carbon dioxide.

1.3.2 Powering systems

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- Mains electricity is delivered to homes and businesses via the National Grid using underground or overhead cables via pylons.
- Energy from solar and wind farms also contributes to the National Grid.
- Batteries store electrical power. Battery capacity varies depending on the number of cells and the voltage within each cell.
- Battery technology continues to evolve – the size of batteries has significantly been reduced but the capacity to power a device has increased.

1.3.3 Choosing appropriate energy sources

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Using renewable energy instead of burning fossil fuels is seen by many as the cleaner and greener way forward. However, there is still an environmental cost:

- Many people consider wind farms and solar panels unsightly and noisy.
- There are concerns that birds could be harmed by wind turbine blades.

Compact renewable energy sources have been developed to be used on portable devices:

- Small solar PV panels can produce a small current to recharge a battery. Flexible solar PV panels can be found on clothing and bags, which can charge a mobile phone.
- Low-powered products can be charged from a small wind generator.
- Clockwork wind-up mechanisms can provide a temporary source of power for mechanical or electronic products.
- Technology allows for electrical devices to be far more efficient in terms of energy used. For example, using **light-emitting diode (LED)** lights instead of filament lamps reduces energy consumption.
- Domestic appliances carry an energy rating label. A+++ is the most efficient; while G is the least.
- Standard connections allow products to be sold worldwide, for example USB slots on computers.
- Rechargeable products are often seen as an advantage by consumers.

Light-emitting diode (LED): an output device that produces light when current flows from anode to cathode; they are more energy efficient than traditional filament lightbulbs.



Figure 1.5 Electronic road sign powered by solar energy

Now test yourself

TESTED

1 Biomass fuels are said to be carbon neutral. Explain the meaning of this term. (2)
2 Describe in detail **one** advantage and **one** disadvantage of wind power. (2)
3 Discuss the disadvantages of using fossil fuels to provide energy. (4)
4 State the correct name for the cells found on solar panels. (1)
5 Explain how technology has impacted the development of batteries. (2)

Exam tip

Make sure you fully understand the meaning of key words and terminology. This will help you to understand the context of each question and enable you to write a full and detailed answer.

Typical mistake

Read exam questions carefully and make sure you fully understand what the question is asking before attempting an answer. The main point of the question can often be missed if an answer is rushed.